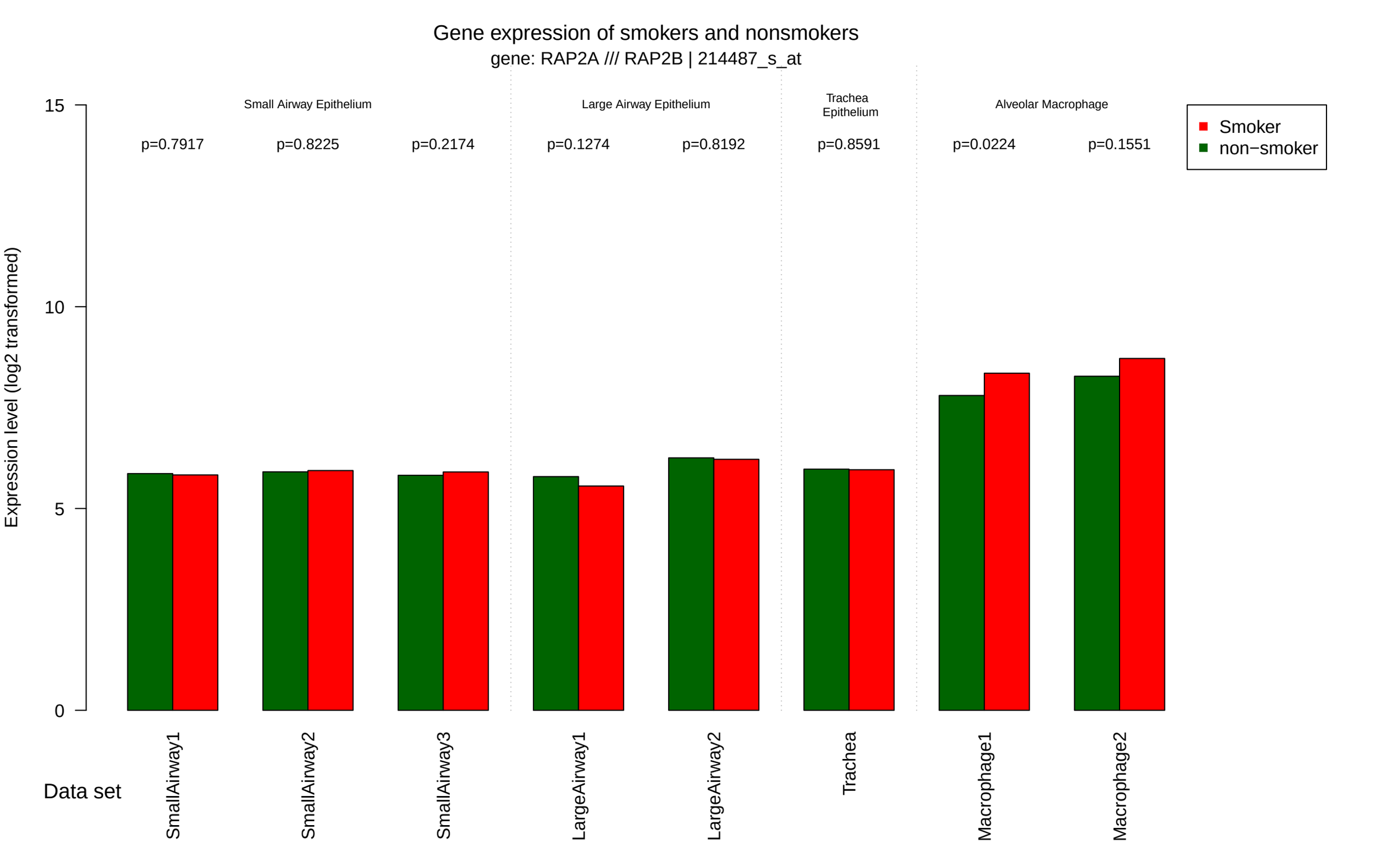
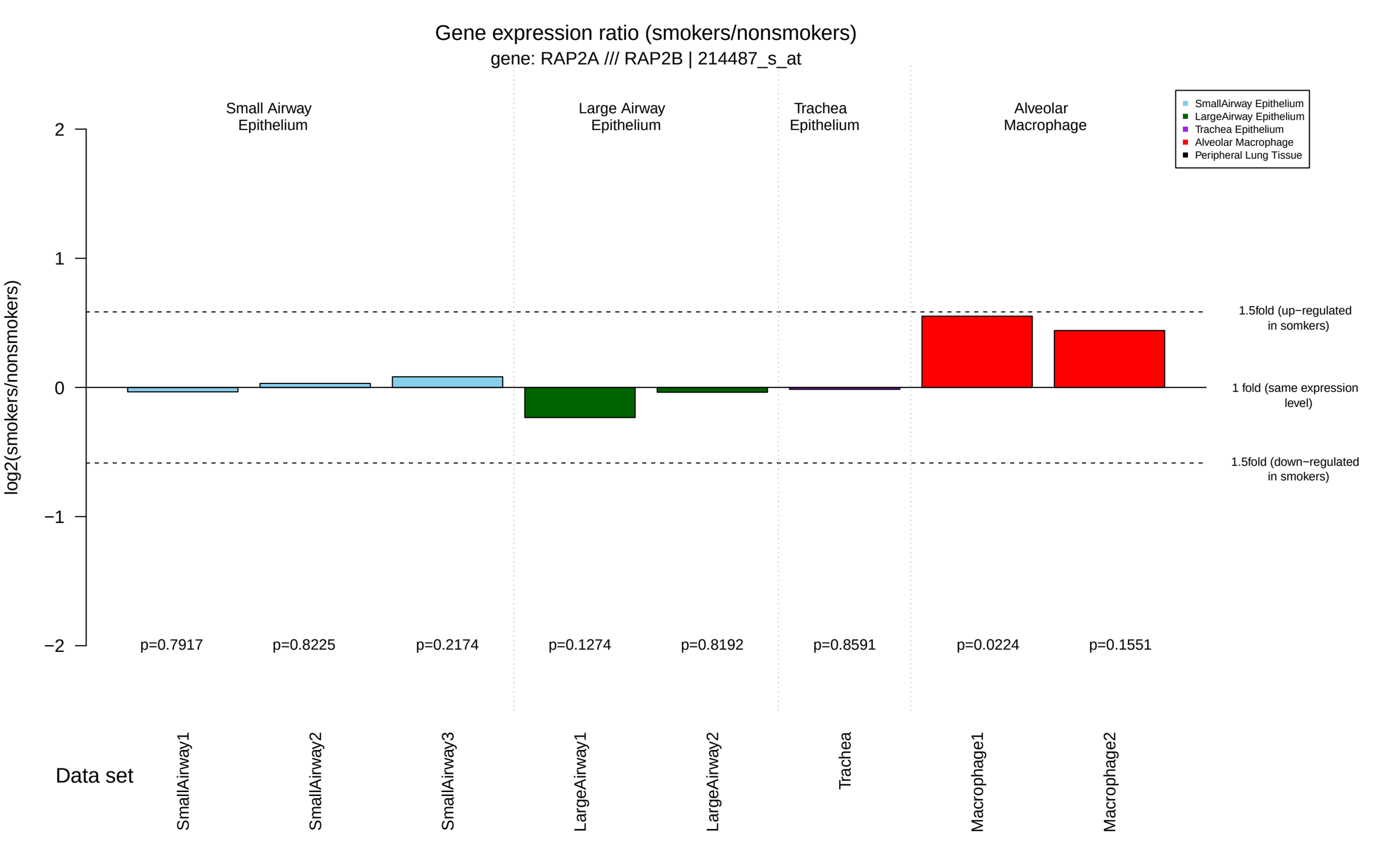
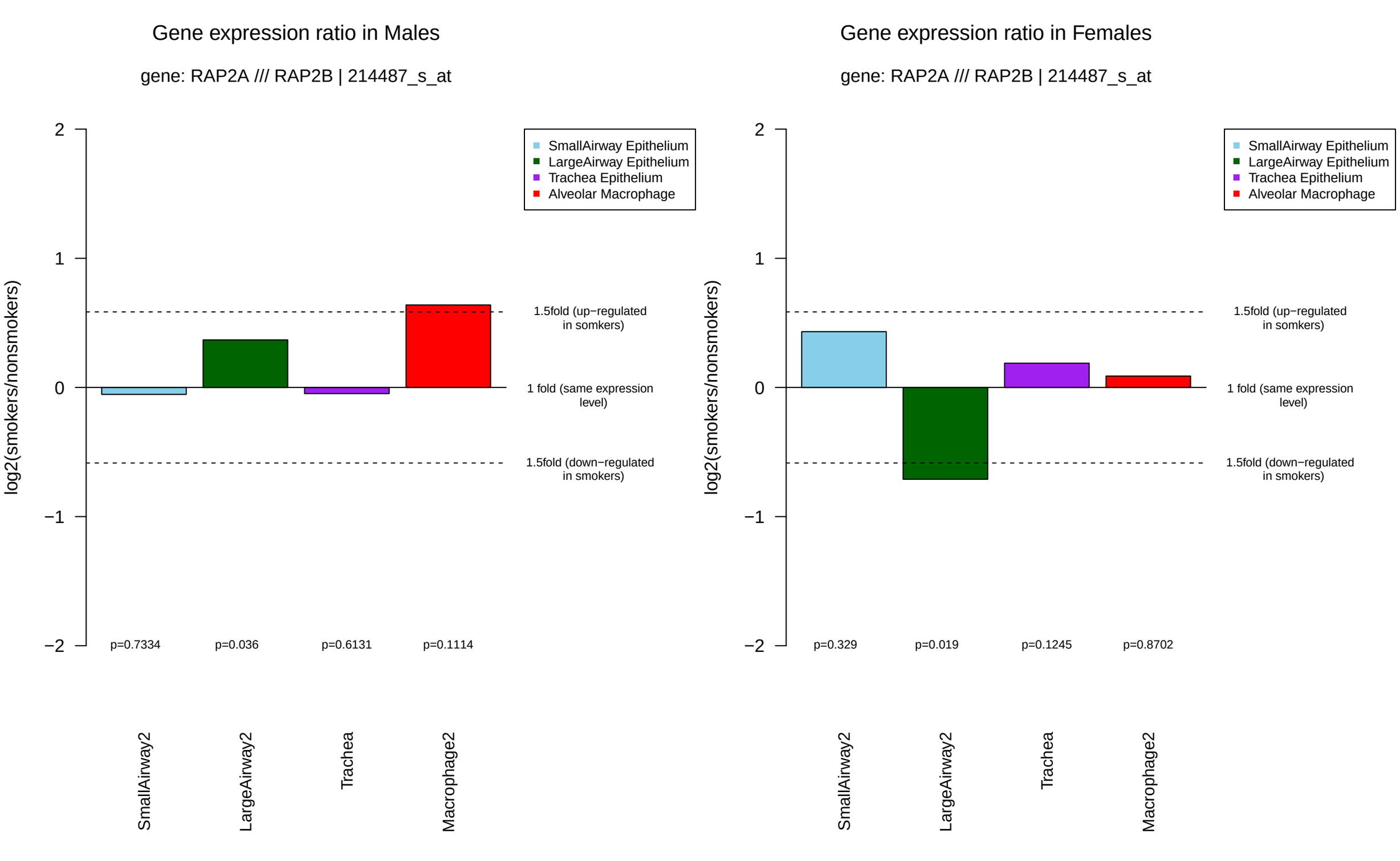
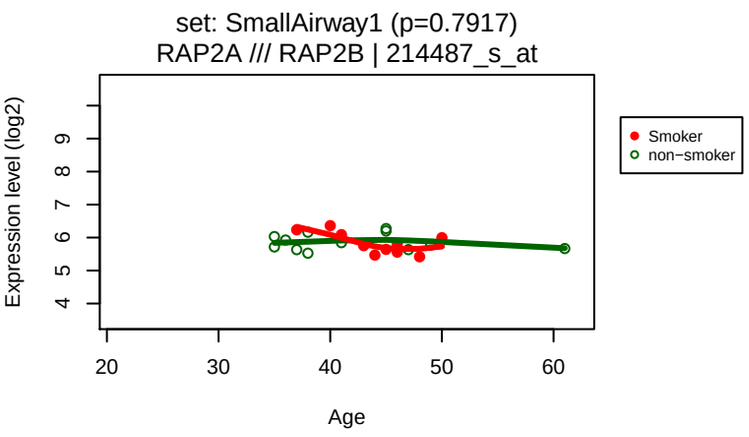
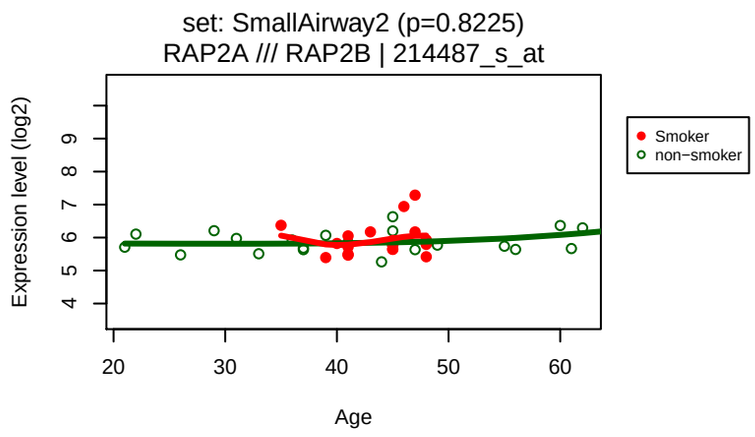
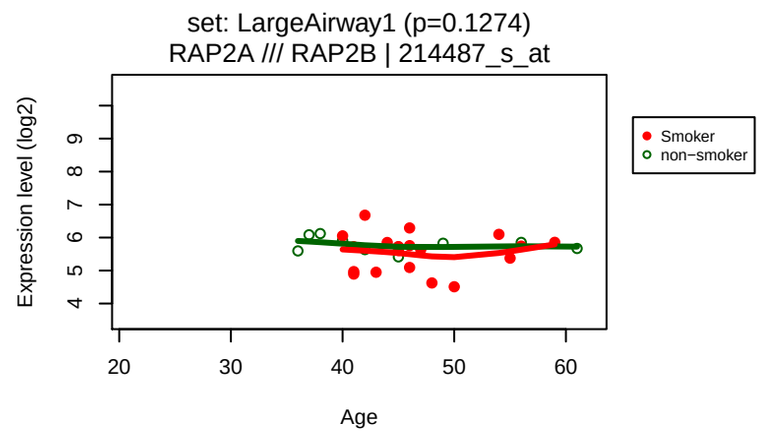

Gene expression of smokers and nonsmokers
gene: RAP2A /// RAP2B | 214487_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7917
p=0.8225
p=0.2174
p=0.1274
p=0.8192
p=0.8591
p=0.0224
p=0.1551
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

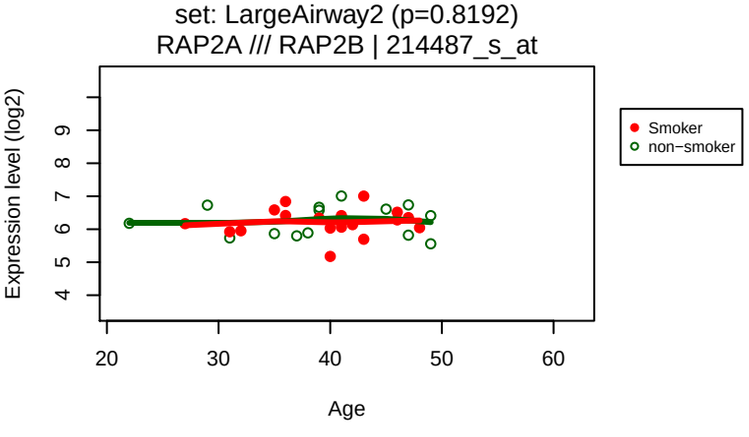

Gene expression ratio (smokers/nonsmokers)
gene: RAP2A /// RAP2B | 214487_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7917
p=0.8225
p=0.2174
p=0.1274
p=0.8192
p=0.8591
p=0.0224
p=0.1551
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

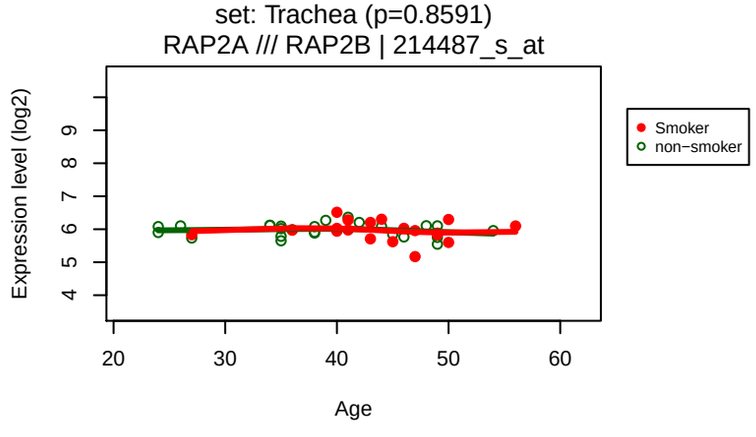

Gene expression ratio in Males
gene: RAP2A /// RAP2B | 214487_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7334
p=0.036
p=0.6131
p=0.1114
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAP2A /// RAP2B | 214487_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.329
p=0.019
p=0.1245
p=0.8702
SmallAirway2
LargeAirway2
Trachea
Macrophage2

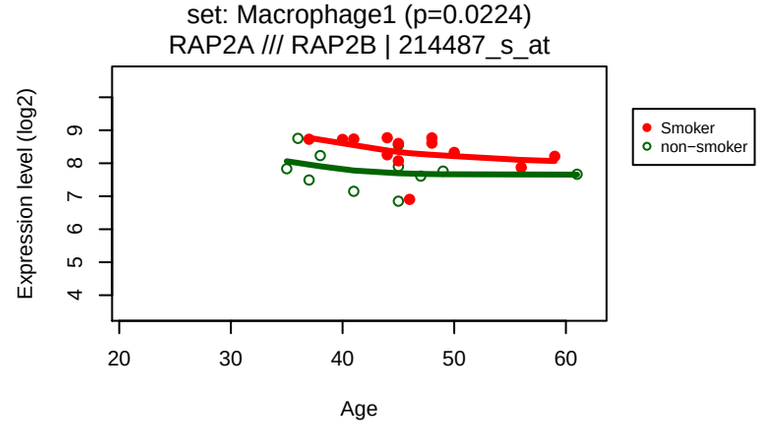

set: SmallAirway1 (p=0.7917)
RAP2A /// RAP2B | 214487_s_at
Expression level (log2)
Age
Smoker
non−smoker

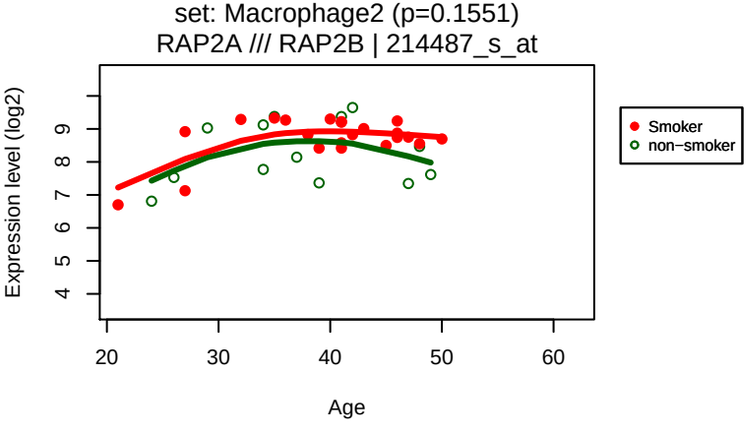

set: SmallAirway2 (p=0.8225)
RAP2A /// RAP2B | 214487_s_at
Expression level (log2)
Age
Smoker
non−smoker

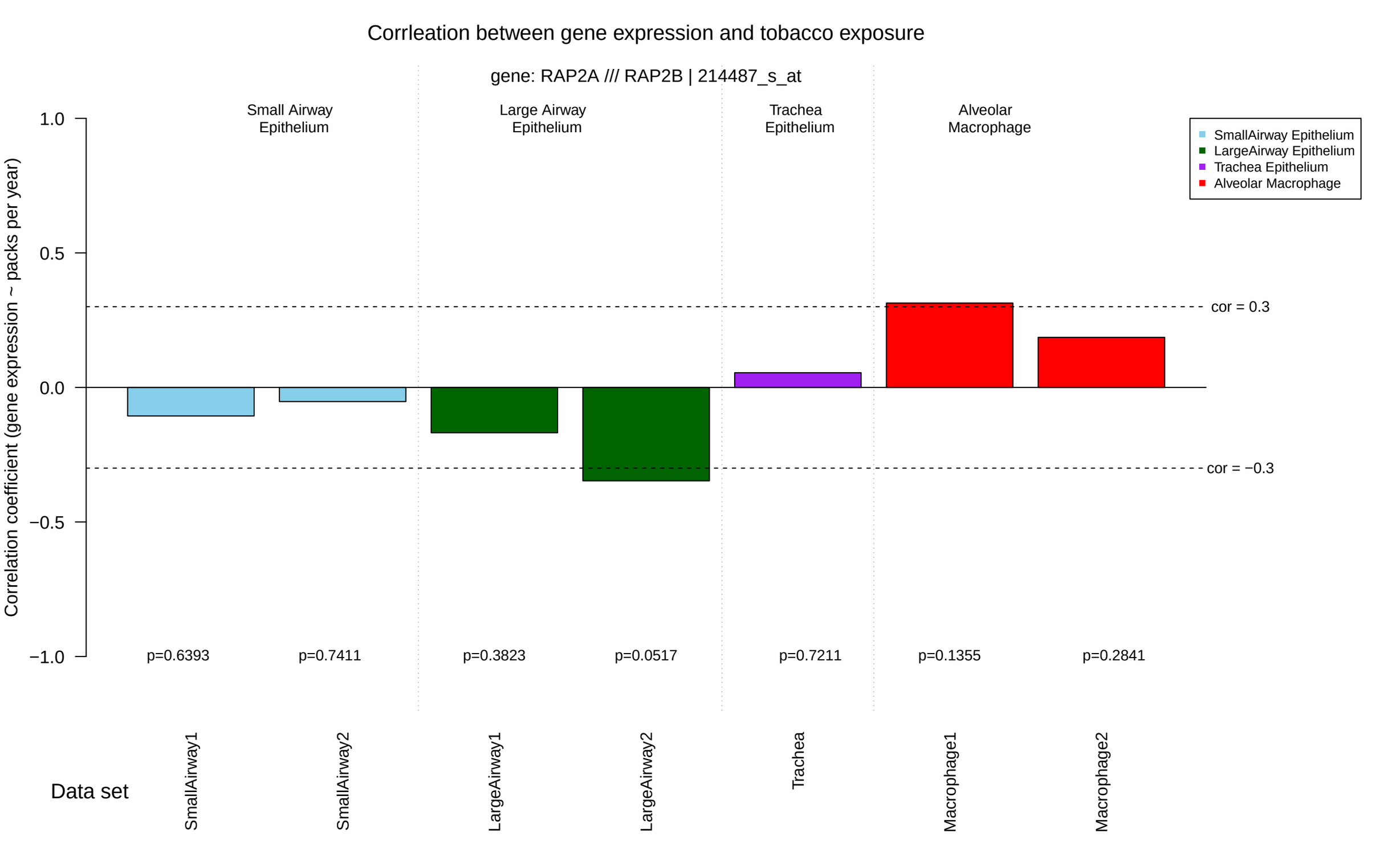

set: LargeAirway1 (p=0.1274)
RAP2A /// RAP2B | 214487_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8192)
RAP2A /// RAP2B | 214487_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8591)
RAP2A /// RAP2B | 214487_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0224)
RAP2A /// RAP2B | 214487_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1551)
RAP2A /// RAP2B | 214487_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAP2A /// RAP2B | 214487_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6393
p=0.7411
p=0.3823
p=0.0517
p=0.7211
p=0.1355
p=0.2841
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2